



Structure Query Language

Queries

STRUCTURE QUERY LANGUAGE

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✅ 100 SQL Problems With Answers

A) BASIC SELECT (1–20)

1. Display all employee details.

```
SELECT * FROM EMP;
```

2. Display all department details.

```
SELECT * FROM DEPT;
```

3. Display employee name and job.

```
SELECT ENAME, JOB FROM EMP;
```

4. Display employee name and salary.

```
SELECT ENAME, SAL FROM EMP;
```

5. Display unique job types.

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```
SELECT DISTINCT JOB FROM EMP;
```

6. Display employees working in department 10.

```
SELECT * FROM EMP WHERE DEPTNO = 10;
```

7. Display managers only.

```
SELECT * FROM EMP WHERE JOB = 'MANAGER';
```

8. Display employees with salary > 2000.

```
SELECT * FROM EMP WHERE SAL > 2000;
```

9. Display employees with salary between 1000 and 3000.

```
SELECT * FROM EMP WHERE SAL BETWEEN 1000 AND 3000;
```

10. Display employees whose name starts with ‘S’.

```
SELECT * FROM EMP WHERE ENAME LIKE 'S%';
```

11. Display employees whose name ends with 'N'.

```
SELECT * FROM EMP WHERE ENAME LIKE '%N';
```

12. Display employees with no commission.

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```
SELECT * FROM EMP WHERE COMM IS NULL;
```

13. Display employees with commission.

```
SELECT * FROM EMP WHERE COMM IS NOT NULL;
```

14. Display employees in ascending order of salary.

```
SELECT * FROM EMP ORDER BY SAL;
```

15. Display employees in descending order of salary.

```
SELECT * FROM EMP ORDER BY SAL DESC;
```

16. Display employees ordered by department then salary.

```
SELECT * FROM EMP ORDER BY DEPTNO, SAL DESC;
```

17. Display employees of dept 10 and 20.

```
SELECT * FROM EMP WHERE DEPTNO IN (10, 20);
```

18. Display first 5 employees.

```
SELECT * FROM EMP LIMIT 5;
```

19. Display employees with job = CLERK.

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```
SELECT * FROM EMP WHERE JOB = 'CLERK';
```

20. Display name and annual salary.

```
SELECT ENAME, SAL*12 AS ANNUAL_SAL FROM EMP;
```

B) AGGREGATE + GROUP BY (21–40)

21. Count total employees.

```
SELECT COUNT(*) FROM EMP;
```

22. Find total salary paid.

```
SELECT SUM(SAL) FROM EMP;
```

23. Find average salary.

```
SELECT AVG(SAL) FROM EMP;
```

24. Find highest salary.

```
SELECT MAX(SAL) FROM EMP;
```

25. Find lowest salary.

```
SELECT MIN(SAL) FROM EMP;
```

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26. Find count of employees in each department.

```
SELECT DEPTNO, COUNT(*) FROM EMP GROUP BY DEPTNO;
```

27. Find average salary for each job.

```
SELECT JOB, AVG(SAL) FROM EMP GROUP BY JOB;
```

28. Find total salary dept-wise.

```
SELECT DEPTNO, SUM(SAL) FROM EMP GROUP BY DEPTNO;
```

29. Dept with more than 3 employees.

```
SELECT DEPTNO, COUNT(*)
FROM EMP
GROUP BY DEPTNO
HAVING COUNT(*) > 3;
```

30. Job with avg salary > 2000.

```
SELECT JOB, AVG(SAL)
FROM EMP
GROUP BY JOB
HAVING AVG(SAL) > 2000;
```

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31. Dept with salary sum > 10000.

```
SELECT DEPTNO, SUM(SAL)
FROM EMP
GROUP BY DEPTNO
HAVING SUM(SAL) > 10000;
```

32. Count employees for each job.

```
SELECT JOB, COUNT(*) FROM EMP GROUP BY JOB;
```

33. Show dept-wise max salary.

```
SELECT DEPTNO, MAX(SAL) FROM EMP GROUP BY DEPTNO;
```

34. Show dept-wise min salary.

```
SELECT DEPTNO, MIN(SAL) FROM EMP GROUP BY DEPTNO;
```

35. Show job-wise salary range.

```
SELECT JOB, MAX(SAL)-MIN(SAL) AS RANGE FROM EMP GROUP BY JOB;
```

36. Jobs having more than 2 employees.

```
SELECT JOB, COUNT(*) FROM EMP GROUP BY JOB HAVING COUNT(*) > 2;
```

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37. Dept where avg comm is not null.

```
SELECT DEPTNO, AVG(COMM) FROM EMP GROUP BY DEPTNO;
```

38. Count employees who have commission per dept.

```
SELECT DEPTNO, COUNT(*) FROM EMP WHERE COMM IS NOT NULL GROUP BY DEPTNO;
```

39. Highest salary person in each department.

```
SELECT DEPTNO, MAX(SAL) FROM EMP GROUP BY DEPTNO;
```

40. Lowest salary per job.

```
SELECT JOB, MIN(SAL) FROM EMP GROUP BY JOB;
```

C) JOINS (41–50)

41. Show employee name and department name.

```
SELECT ENAME, DNAME

FROM EMP e

JOIN DEPT d ON e.DEPTNO = d.DEPTNO;
```

42. Show employee name, job, department, and location.

```
SELECT ENAME, JOB, DNAME, LOC

FROM EMP e
```

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```
JOIN DEPT d USING (DEPTNO);
```

43. Show employee & manager name.

```
SELECT e.ENAME AS EMPLOYEE, m.ENAME AS MANAGER

FROM EMP e

JOIN EMP m ON e.MGR = m.EMPNO;
```

44. Show employee salary grade.

```
SELECT ENAME, SAL, GRADE

FROM EMP e

JOIN SALGRADE s ON e.SAL BETWEEN s.LOSAL AND s.HISAL;
```

45. List all combinations of EMP x DEPT (Cartesian).

```
SELECT * FROM EMP CROSS JOIN DEPT;
```

46. Employees with department locations.

```
SELECT ENAME, LOC FROM EMP e JOIN DEPT d USING(DEPTNO);
```

47. Left join all employees even if no dept.

```
SELECT ENAME, DNAME

FROM EMP e LEFT JOIN DEPT d USING(DEPTNO);
```

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48. Right join show all departments.

```
SELECT ENAME, DNAME FROM EMP e RIGHT JOIN DEPT d USING(DEPTNO);
```

49. Employee, department & salary grade.

```
SELECT ENAME, DNAME, GRADE
FROM EMP e
JOIN DEPT d USING(DEPTNO)
JOIN SALGRADE s ON SAL BETWEEN LOSAL AND HISAL;
```

50. Count employees dept-wise with name of dept.

```
SELECT DNAME, COUNT(*)
FROM EMP e
JOIN DEPT d USING(DEPTNO)
GROUP BY DNAME;
```

51. Display employees who earn between 2000 and 4000

```
SELECT * FROM EMP
WHERE SAL BETWEEN 2000 AND 4000;
```

52. Display employees whose commission is NULL

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```
SELECT * FROM EMP

WHERE COMM IS NULL;
```

53. Display employees whose commission is NOT NULL

```
SELECT * FROM EMP

WHERE COMM IS NOT NULL;
```

54. Display employees whose name starts with 'S'

```
SELECT * FROM EMP

WHERE ENAME LIKE 'S%';
```

55. Display employees whose name ends with 'N'

```
SELECT * FROM EMP

WHERE ENAME LIKE '%N';
```

56. Display employees whose name contains 'A'

```
SELECT * FROM EMP

WHERE ENAME LIKE '%A%';
```

57. Display job-wise total salary

```
SELECT JOB, SUM(SAL) AS TOTAL_SALARY
```

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```
FROM EMP

GROUP BY JOB;
```

58. Display department-wise employee count

```
SELECT DEPTNO, COUNT(*) AS EMP_COUNT

FROM EMP

GROUP BY DEPTNO;
```

59. Display employees hired in the year 1981

```
SELECT * FROM EMP

WHERE EXTRACT(YEAR FROM HIREDATE) = 1981;
```

60. Display employees with salary greater than the company's average salary

```
SELECT * FROM EMP

WHERE SAL > (SELECT AVG(SAL) FROM EMP);
```

D) SUBQUERIES (61–80)

61. Employees with salary greater than average.

```
SELECT * FROM EMP WHERE SAL > (SELECT AVG(SAL) FROM EMP);
```

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62. Employee(s) with highest salary.

```
SELECT * FROM EMP WHERE SAL = (SELECT MAX(SAL) FROM EMP);
```

63. Employee(s) with lowest salary.

```
SELECT * FROM EMP WHERE SAL = (SELECT MIN(SAL) FROM EMP);
```

64. Employees working in SALES dept.

```
SELECT ENAME

FROM EMP

WHERE DEPTNO = (SELECT DEPTNO FROM DEPT WHERE DNAME='SALES');
```

65. Employees with same salary as SCOTT.

```
SELECT * FROM EMP WHERE SAL = (SELECT SAL FROM EMP WHERE ENAME='SCOTT');
```

66. Employees earning more than FORD.

```
SELECT ENAME FROM EMP WHERE SAL > (SELECT SAL FROM EMP WHERE ENAME='FORD');
```

67. Employees in same dept as MILLER.

```
SELECT ENAME FROM EMP

WHERE DEPTNO = (SELECT DEPTNO FROM EMP WHERE ENAME='MILLER');
```

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68. Dept with minimum salary employee.

```
SELECT DEPTNO

FROM EMP

WHERE SAL = (SELECT MIN(SAL) FROM EMP);
```

69. Employees not managers.

```
SELECT ENAME

FROM EMP

WHERE EMPNO NOT IN (SELECT DISTINCT MGR FROM EMP WHERE MGR IS NOT NULL);
```

70. Employees whose salary equals dept-wise max.

```
SELECT *

FROM EMP

WHERE (SAL, DEPTNO) IN

  (SELECT MAX(SAL), DEPTNO FROM EMP GROUP BY DEPTNO);
```

71. Employee working in location = NEW YORK.

```
SELECT ENAME

FROM EMP

WHERE DEPTNO IN (SELECT DEPTNO FROM DEPT WHERE LOC='NEW YORK');
```

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72. Employees who get commission more than avg commission.

```
SELECT *

FROM EMP

WHERE COMM > (SELECT AVG(COMM) FROM EMP);
```

73. Jobs in dept 10.

```
SELECT DISTINCT JOB FROM EMP WHERE DEPTNO=10;
```

74. Employees whose job is same as CLARK's job.

```
SELECT ENAME

FROM EMP

WHERE JOB = (SELECT JOB FROM EMP WHERE ENAME='CLARK');
```

75. Employee whose salary is 2nd highest.

```
SELECT * FROM EMP

WHERE SAL = (SELECT MAX(SAL) FROM EMP WHERE SAL < (SELECT MAX(SAL) FROM EMP));
```

76. Display employees who earn more than dept 30 employees.

```
SELECT ENAME

FROM EMP

WHERE SAL > ALL (SELECT SAL FROM EMP WHERE DEPTNO=30);
```

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77. Employees who earn at least one employee of dept 20.

```
SELECT ENAME

FROM EMP

WHERE SAL > ANY (SELECT SAL FROM EMP WHERE DEPTNO=20);
```

78. Employees whose salary is same as someone in dept 10.

```
SELECT ENAME

FROM EMP

WHERE SAL IN (SELECT SAL FROM EMP WHERE DEPTNO=10);
```

79. Identify manager of ALLEN.

```
SELECT ENAME

FROM EMP

WHERE EMPNO = (SELECT MGR FROM EMP WHERE ENAME='ALLEN');
```

80. Employees with salary between dept min and max.

```
SELECT *

FROM EMP

WHERE SAL BETWEEN (SELECT MIN(SAL) FROM EMP)

AND (SELECT MAX(SAL) FROM EMP);
```

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E) DDL Problems (81–90)

81. Create student table.

```
CREATE TABLE STUDENT (
  ROLL INT PRIMARY KEY,
  NAME VARCHAR(30)
);
```

82. Create employee table.

```
CREATE TABLE EMPLOYEE (
  ID INT,
  NAME VARCHAR(20),
  SAL FLOAT
);
```

83. Create database SCHOOL.

```
CREATE DATABASE SCHOOL;
```

84. Drop student table.

```
DROP TABLE STUDENT;
```

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85. Add column AGE in EMP.

```
ALTER TABLE EMP ADD AGE INT;
```

86. Modify ENAME length to 50.

```
ALTER TABLE EMP MODIFY ENAME VARCHAR(50);
```

87. Drop AGE column.

```
ALTER TABLE EMP DROP COLUMN AGE;
```

88. Rename EMP to EMPLOYEE_DATA.

```
RENAME TABLE EMP TO EMPLOYEE_DATA;
```

89. Remove all rows from EMP.

```
TRUNCATE TABLE EMP;
```

90. Create a view of high salary employees.

```
CREATE VIEW HIGH_SAL_EMP AS

SELECT * FROM EMP WHERE SAL > 3000;
```

F) DML Problems (91–100)

91. Insert employee record.

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```
INSERT INTO EMP VALUES (1001,'RAHUL','CLERK',7902,'2023-03-12',1500,NULL,20);
```

92. Insert department record.

```
INSERT INTO DEPT VALUES (50,'TECH','MUMBAI');
```

93. Increase salary of clerks by 500.

```
UPDATE EMP SET SAL = SAL + 500 WHERE JOB='CLERK';
```

94. Delete employee with ID 1001.

```
DELETE FROM EMP WHERE EMPNO = 1001;
```

95. Change department 30 location to DELHI.

```
UPDATE DEPT SET LOC='DELHI' WHERE DEPTNO=30;
```

96. Insert new student.

```
INSERT INTO STUDENT VALUES (1,'SOHAM');
```

97. Update student name.

```
UPDATE STUDENT SET NAME='PATIL' WHERE ROLL=1;
```

98. Delete student.

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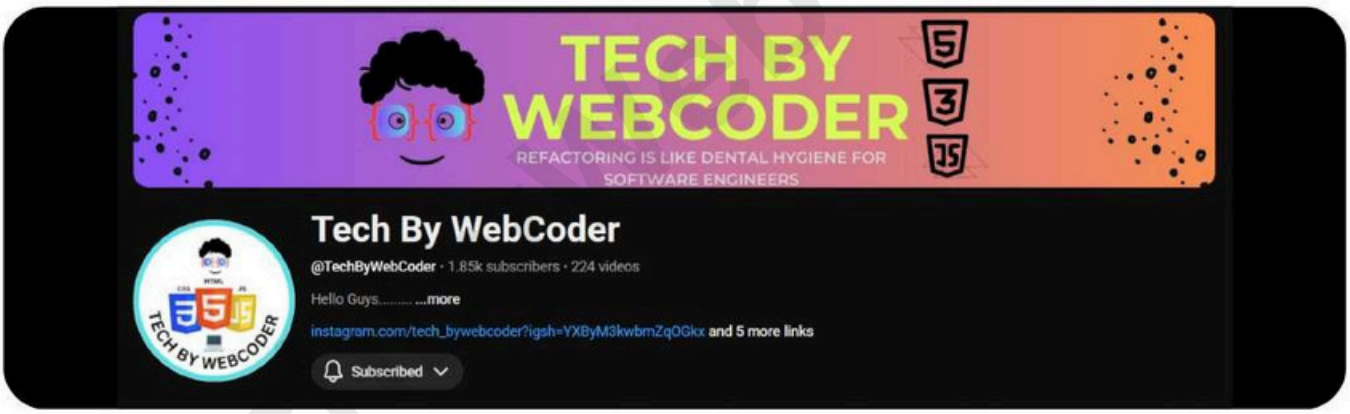
```
DELETE FROM STUDENT WHERE ROLL=1;
```

99. Insert new employee with commission.

```
INSERT INTO EMP VALUES (2001,'AMIT','SALESMAN',7698,'2024-02-05',1600,500,30);
```

100. Update commission for employees of dept 30.

```
UPDATE EMP SET COMM = 300 WHERE DEPTNO = 30;
```



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Thank you !

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